Retail Sales Analysis & Business Insights

Overview

This project analyzes a global superstore dataset using SQL to understand how the business is performing. The analysis focuses on sales trends over time, top-performing regions and cities, best-selling products, customer behavior, and shipping performance.

The results show that a few regions and product categories generate most of the revenue, while some cities and products perform poorly. Customer segments also have different spending patterns.These insights help the business improve inventory planning, focus on high-value customers, and make better decisions to increase sales and overall performance.

PROBLEM STATEMENT

The company does not have a clear understanding of its sales performance, customer behavior, and business operations. It is difficult to know which regions, cities, and products are performing well and how sales change over time.

Key challenges include identifying top-performing areas, understanding product performance, analyzing customer segments, and evaluating the impact of shipping on business performance.

What's Happening?

● The company has sales data, but it is not properly analyzed.

● The data is not well organized, making it hard to get useful insights.

● There is no clear idea of which regions, cities, or products generate the most sales.

● Customer behavior and buying patterns are not clearly understood.

Solution Goal

● To clean and organize the dataset into a proper structure.

● To use SQL to analyze sales, customers, and products.

● To find useful insights that help improve sales and business decisions.

impact is not clearly evaluated.

Because of these gaps, the business may face problems such as poor inventory management, ineffective marketing strategies, and missed opportunities for growth. Therefore, there is a need for a structured data analysis approach to gain meaningful insights and support better business decisions.

Project Objectives

The main goals of this project are:

- To understand overall sales and how they change each year
- To find which regions and cities perform the best
- To check which product categories and sub-categories bring more sales
- To identify important customers and understand different customer groups
- To study how shipping methods affect business performance
- To find low-performing areas and possible growth opportunities

DATA DESCRIPTION

- **Title:** Global Superstore Sales Dataset
- **Sample Size:** ~9800 Rows | 15 Columns
- **Data Type:** Mixed — Numeric, Text, and Categorical
- **Table Used:** Superstore Sales Table

Key Fields:

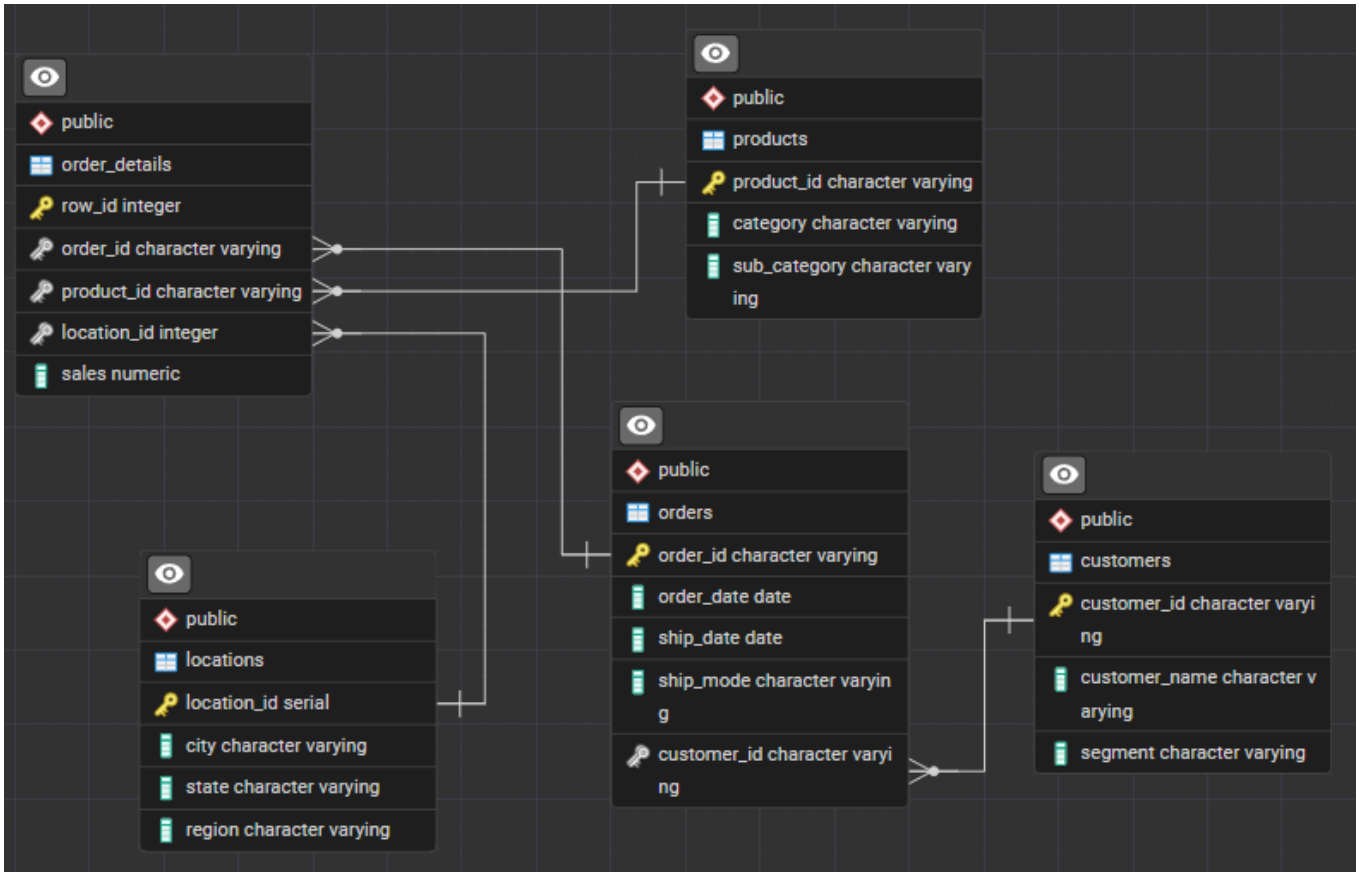
- row_id, order_id, order_date, ship_date
- customer_id, customer_name, segment
- city, state, region
- product_id, category, sub_category
- sales, ship_mode

Data Cleaning and Preparation

The dataset used in this project was already mostly clean and well-structured. However, some basic preparation steps were performed before starting the analysis.

- Column names were standardized to make them clear, consistent, and easy to use in queries
- Checked for missing (null) values to ensure data completeness
- Checked for duplicate records to avoid incorrect results
- Unnecessary spaces and formatting issues in column names were corrected
- The date column was used to extract the year for time-based analysis
- Data types were verified to ensure correct usage (such as dates and numeric values)
- The dataset was reviewed to understand relationships between fields like sales, customers, and products

These steps ensured that the data was accurate, consistent, and ready for analysis using SQL.



The data model is organized into multiple tables such as **orders, order_details, customers, products, and locations**.

Each order in the **orders** table can have multiple entries in the **order_details** table, creating a **one-to-many relationship**. The **order_id** connects both tables.

The **products** table stores product information and is linked to order_details using **product_id**. The **customers** table is connected to orders using **customer_id** to track customer activity.

The **locations** table contains city, state, and region details, helping in location-based analysis.

Overall, this structure keeps the data organized, reduces duplication, and makes analysis easier.

1.Total sales (overall)

```
select round(sum(sales)) as total_sales
from order_details;
```

	total_sales numeric
1	2261537

Insight

This shows the overall revenue generated from all orders in the dataset. It gives a clear idea of the **business size and performance level** over the given time period.

2.What is the yearly trend of total sales over time?

```sql
with yearly as (
  select
    extract(year from o.order_date) as year,
    sum(od.sales) as total_sales
  from order_details od
  join orders o on od.order_id = o.order_id
  group by year
)
select
  year,
  total_sales,
  coalesce(lag(total_sales) over(order by year), 0) as prev_year,
  case
    when lag(total_sales) over(order by year) is null then 0
    else round(
      (total_sales - lag(total_sales) over(order by year)) * 100.0 /
      lag(total_sales) over(order by year),
      2
    )
  end as growth_pct
from yearly
order by year;
```

	year numeric	total_sales numeric	prev_year numeric	growth_pct numeric
1	2015	479856.2081	0	0
2	2016	459436.0054	479856.2081	-4.26
3	2017	600192.5500	459436.0054	30.64
4	2018	722052.0192	600192.5500	20.30

Insight

- Sales dropped slightly in 2016 compared to 2015

- After that, sales increased strongly in 2017 and 2018
- The business shows **good growth in the last two years.**

3. How many orders were placed each year?

```sql
with yearly as (
  select
    extract(year from order_date) as year,
    count(*) as total_orders
  from orders
  group by year
)
select
  year,
  total_orders,
  coalesce(lag(total_orders) over(order by year), 0) as prev_orders,
  coalesce(
    round(
      (total_orders - lag(total_orders) over(order by year)) * 100.0 /
      lag(total_orders) over(order by year),
      2
    ), 0
  ) as growth_pct
from yearly
order by year;
```

	year numeric	total_orders bigint	prev_orders bigint	growth_pct numeric
1	2015	947	0	0
2	2016	1019	947	7.60
3	2017	1295	1019	27.09
4	2018	1661	1295	28.26

Insight

- The number of orders is **increasing every year**
- Growth is strong, especially in 2017 and 2018
- More customers are buying over time

4. Which region generates the highest revenue?

```
select l.region, round(sum(od.sales)) as total_sales
from order_details od
join locations l on od.location_id = l.location_id
group by l.region
order by total_sales desc;
```

	region character varying	total_sales numeric
1	West	710220
2	East	669519
3	Central	492647
4	South	389151

Insight

- The **West region performs the best** and generates the most sales
- The **South region performs the worst** with the lowest sales
- Sales are not equal across regions

5. Which cities generate the highest total sales?

```sql
select
  l.city,
  round(sum(od.sales), 2) as total_sales
from order_details od
join locations l on od.location_id = l.location_id
group by l.city
order by total_sales desc
limit 5;
```

	city character varying	total_sales numeric
1	New York City	252462.55
2	Los Angeles	173420.18
3	Seattle	116106.32
4	San Francisco	109041.12
5	Philadelphia	108841.75

Insight

- **New York City generates the most sales**
- Sales are mainly coming from a few top cities
- Big cities contribute more to revenue

6. How do different shipping modes perform in terms of order volume, delivery time, and revenue?

```sql
select
  o.ship_mode,
  count(distinct o.order_id) as total_orders,
  round(avg(o.ship_date - o.order_date), 2) as avg_delivery_days,
  round(sum(od.sales), 2) as total_sales
from orders o
join order_details od on o.order_id = od.order_id
group by o.ship_mode;
```

	ship_mode character varying	total_orders bigint	avg_delivery_days numeric	total_sales numeric
1	First Class	772	6.59	345572.26
2	Same Day	261	0.57	125219.04
3	Second Class	944	4.70	449914.18
4	Standard Class	2945	12.16	1340831.31

Insight

- **Standard Class is used the most** and brings the highest sales
- **Same Day delivery is fastest**, but used very less
- Faster shipping has fewer orders, slower shipping has more orders
- Customers mostly prefer **cheaper shipping options**

7. Which product category drives most sales?

```sql
select p.category, round(sum(od.sales)) as total_sales
from products p
join order_details od
on p.product_id = od.product_id
group by p.category
order by total_sales desc;
```

	category character varying	total_sales numeric
1	Technology	827456
2	Furniture	728659
3	Office Supplies	705422

Insight

- **Technology is the top-selling category**
- Furniture and Office Supplies have similar sales
- Technology products bring more revenue than other categories

8.Which product sub category drives most sales?

```sql
select p.sub_category, round(sum(od.sales)) as total_sales
from products p
join order_details od
on p.product_id = od.product_id
group by p.sub_category
order by total_sales desc;
```

	sub_category character varying	total_sales numeric
1	Phones	327782
2	Chairs	322823
3	Storage	219343
4	Tables	202811
5	Binders	200029
6	Machines	189239
7	Accessories	164187
8	Copiers	146248
9	Bookcases	113813
10	Appliances	104618
11	Furnishings	89212
12	Paper	76828
13	Supplies	46420
14	Art	26705
15	Envelopes	16128
16	Labels	12348
17	Fasteners	3002

Insight

- **Phones and Chairs are the top-selling products**
- A few sub-categories generate most of the sales
- Some products like **Fasteners, Labels, Envelopes** have very low sales
- Sales are not equal across all products

9.Which sub-category products are purchased the most based on order count?

```
select
  p.sub_category,
  count(*) as total_orders
from order_details od
join products p on od.product_id = p.product_id
group by p.sub_category
order by total_orders desc;
```

	sub_category character varying	total_orders bigint
1	Binders	1492
2	Paper	1338
3	Furnishings	931
4	Phones	876
5	Storage	832
6	Art	785
7	Accessories	756
8	Chairs	607
9	Appliances	459
10	Labels	357
11	Tables	314
12	Envelopes	248
13	Bookcases	226
14	Fasteners	214
15	Supplies	184
16	Machines	115
17	Copiers	66

Insight

- **Binders and Paper are ordered the most**
- These are low-cost, frequently used items
- Some products like **Copiers and Machines** have very few orders
- High orders do not always mean high sales value

10.Which sub-category generates the highest sales within each category?

```sql
with base as (
  select
    p.category,
    p.sub_category,
    round(sum(od.sales)) as total_sales
  from order_details od
  join products p
    on p.product_id = od.product_id
  group by p.category, p.sub_category
),
ranked as (
  select *,
    rank() over (
      partition by category
      order by total_sales desc
    ) as rnk
  from base
)
select
  category,
  sub_category,
  total_sales
from ranked
where rnk <= 2;
```

	category character varying	sub_category character varying	total_sales numeric
1	Furniture	Chairs	322823
2	Furniture	Tables	202811
3	Office Supplies	Storage	219343
4	Office Supplies	Binders	200029
5	Technology	Phones	327782
6	Technology	Machines	189239

Insight

- Each category has **2 main products that bring most of the sales**
- **Phones and Chairs are strong performers**
- Sales are mainly driven by a few key sub-categories

11. Who are the most valuable customers?

```sql
with cte as
(select
o.customer_id, sum(od.sales) as total_sales
from order_details od
join orders o on od.order_id = o.order_id
group by o.customer_id
)
select customer_name, total_sales
from cte
join customers c on c.customer_id = cte.customer_id
order by total_sales desc
limit 5;
```

	customer_name character varying	total_sales numeric
1	Sean Miller	25043.050
2	Tamara Chand	19052.218
3	Raymond Buch	15117.339
4	Tom Ashbrook	14595.620
5	Adrian Barton	14473.571

Insight

- A few customers generate **high sales value**
- These customers are very important for the business
- Sales are partly dependent on top customers

12.Which customer segment (Consumer, Corporate, Home Office) pays more per order?

```sql
with order_totals as (
  select
    o.order_id,
    c.segment,
    sum(od.sales) as order_value
  from orders o
  join order_details od on o.order_id = od.order_id
  join customers c on o.customer_id = c.customer_id
  group by o.order_id, c.segment
)

select
  segment,
  count(order_id) as total_orders,
  round(avg(order_value), 2) as avg_order_value
from order_totals
group by segment
order by avg_order_value desc;
```

	segment character varying	total_orders bigint	avg_order_value numeric
1	Home Office	894	475.37
2	Corporate	1491	461.77
3	Consumer	2537	452.53

Insights

- Home Office customers spend more money per order
- Consumer customers place the most orders but spend less per order
- Corporate customers make frequent purchases with moderate spending
- Home Office segment shows high-value buying behavior with fewer orders
- Consumer and Corporate segments depend on order volume rather than order value
- Different customer segments have distinct buying patterns and priorities

13.Which product is most frequently sold in the top-performing cities?

```sql
with top_cities as (
select
l.city,
sum(od.sales) as city_sales
from order_details od
join locations l on od.location_id = l.location_id
group by l.city
order by city_sales desc
limit 5
),
city_products as (
select
l.city,
p.sub_category,
sum(od.sales) as total_sales,
row_number() over (
partition by l.city
order by sum(od.sales) desc
) as rn
from order_details od
join products p on od.product_id = p.product_id
join locations l on od.location_id = l.location_id
where l.city in (select city from top_cities)
group by l.city, p.sub_category
)
select
cp.city,
cp.sub_category,
cp.total_sales
from city_products cp
join top_cities tc on cp.city = tc.city
where cp.rn = 1
order by tc.city_sales desc;
```

	city character varying	sub_category character varying	total_sales numeric
1	New York City	Chairs	39576.519
2	Los Angeles	Phones	28981.488
3	Seattle	Binders	17700.152
4	San Francisco	Supplies	13586.39
5	Philadelphia	Phones	19432.182

Insight

- **New York City is the highest sales city, where Chairs contribute the most sales, showing strong demand for furniture products in this city**
- Los Angeles also performs well, with Phones generating the highest sales, indicating high demand for technology products
- Customer demand is **different in each city**

14.Which cities are underperforming and which sub-category products least sales in those cities?

```sql
with low_cities as (
select
l.city,
sum(od.sales) as city_sales
from order_details od
join locations l on od.location_id = l.location_id
group by l.city
order by city_sales asc
limit 5
),
city_products as (
select
l.city,
p.sub_category,
sum(od.sales) as total_sales,
row_number() over (
partition by l.city
order by sum(od.sales) asc
) as rn
from order_details od
join products p on od.product_id = p.product_id
join locations l on od.location_id = l.location_id
where l.city in (select city from low_cities)
group by l.city, p.sub_category
)
select
cp.city,
cp.sub_category,
cp.total_sales
from city_products cp
join low_cities tc on cp.city = tc.city
where cp.rn = 1
order by tc.city_sales asc;
```

	city character varying	sub_category character varying	total_sales numeric
1	Abilene	Appliances	1.392
2	Elyria	Binders	1.824
3	Jupiter	Art	2.064
4	Pensacola	Binders	2.214
5	Ormond Beach	Binders	2.808

Insight

- These cities are low-performing markets, and even the product sales within them are very low
- Some products like **Binders and Art are not selling well** in these cities
- Abilene is the lowest sales occurring city, where Appliances have the lowest sales
- Demand is very low in these locations

Recommendations

Based on the findings, several strategic recommendations can be made.

- The business should focus more on **top-performing cities** to maximize revenue.**New York City generates the highest sales, mainly from Chairs (39,576)**, and **Los Angeles performs strongly with Phones (28,981)**.
- Increasing stock and marketing in these cities can further improve sales.
- The company should follow a **city-specific product strategy** instead of using the same approach everywhere. **Seattle performs well in Binders**, while **San Francisco shows higher demand for Supplies**.
- This shows that customer preferences are different in each city.
- Keep more stock of **high-demand products** in each city
- Give **loyalty rewards** (discounts, early access)
- Offer **free or faster delivery**
- High-performing product categories should be prioritized for stock management and promotional activities, while low-performing products should be re-evaluated for pricing, positioning, or discontinuation.

- Customer segmentation should be leveraged to design personalized marketing campaigns, particularly targeting high-value customers to improve retention and lifetime value.